

football-synthesizer

Broadcast video to validated tactical measurement -- project report

Manchester United, English Premier League 2024-25. 12 matches, 6 opponents, both legs. Prepared for examination, IIIT Delhi, December 2026. Report date: 2026-07-28.

1. Summary

This project turns an ordinary television broadcast into a measured tactical record. The input is a normal match replay: one moving camera, replays, close-ups, crowd shots. The output is player and ball positions on a real 105 x 68 m pitch, tracked events, and a written report that states a number only when the evidence supports it.

Measure exactly what the broadcast supports. Prove where the measurement stops. Report nothing that cannot be grounded in an external check.

Every stage of the pipeline carries an external validation number: an official statistic, a public benchmark scored by its own evaluator, a pre-registered baseline, or a hand-annotated ground truth. When a stage cannot support a claim, the system is built to say so instead of guessing. Twelve of 102 logged claims in this project are retractions of our own earlier work. The retraction trail is a feature of the method, not a defect in it.

The corpus is a repeat-measurement design: 12 Manchester United matches from the 2024-25 season, arranged as six home-and-away pairs against the same six opponents, spanning the change of manager from Erik ten Hag to Ruben Amorim. Playing the same team twice lets us separate "this is Manchester United" from "this is what one opponent did to them on one day".

The rest of this report has six parts: the pipeline stage by stage, the corpus design, the findings on team style, the player-attribution research line, the retraction record, and the plan for the SoccerNet FOOTPASS benchmark leading to December.

Sources. [demo/showcase.html](#); [docs/PROJECT_DEFINITION.md](#); [STATUS.md](#).

2. The pipeline, stage by stage

Six stages. Each stage carries its own external number. Any stage can veto the sentence built downstream of it.

2.1 Video and registry

Full-match replays are supplied by the user and chunked into ten-minute segments. Every match path is looked up from `data/matches.yaml` through a registry module. No file in the codebase hard-codes a path to a match.

2.2 Detect and track

Player detection and multi-object tracking (YOLO plus BoT-SORT), then a jersey-number layer and a re-identification relink on top. Score on the public SoccerNet-GSR validation split, 58 sequences, scored by the official `sn-trackeval` evaluator, not by us:

Table 1. Tracking and identity, GS-HOTA on SoccerNet-GSR valid split (58 sequences)

Arm	GS-HOTA	Note
Pre-jersey baseline	14.76	ident-001
Jersey propagation added	22.85	ident-004; repaired 15/58 regressed sequences down to 1
Tracklet connector added (Stage 1 below)	24.18	+1.33; 51/58 sequences helped

2.3 Pitch metres

A homography is fitted per frame and rejected when its residual error exceeds 1 m, so a player receives a coordinate only when the geometry is trustworthy. Usable geometry covers 26.4% of the whole broadcast on a uniform 1-in-5 frame grid, and 80.5% of live wide-camera frames. The gap between those two numbers is the central honesty problem of the project: a broadcast spends much of its time on replays, close-ups and graphics, and none of that time yields measurement.

2.4 Ball and events

Ball detection, then a linking pass. The only ball-coverage number this project allows itself to quote is the coverage *after* that linking pass: 32-52% of sampled frames across the twelve matches. A pre-link number was once quoted as coverage; that claim was retracted end to end (see Section 6).

Table 2. Event detection, checked against external sources

Event	Result	Checked against
Pass counts per match	0.97-1.09x official on 10 of 12 matches	Official match statistics. Threshold frozen on the first match, never retuned.
Goal detection, zero-shot	13/13 recall, 81.3% precision (16 peaks, 3 false positives)	Six matches with an official goal oracle, including a 0-0 negative control that fires zero peaks. Every true goal is the top-scoring peak in its match.

2.5 Identity

Jersey numbers are read from close-up crops, propagated across re-identification merge groups, and checked against the team sheet. It works, and it does not work often enough: only 1.0-4.2% of Manchester United's attributed passes carry a named player. That single measurement is why this project ships no per-player statistics anywhere.

2.6 Gated report

Team-level metrics are assembled into a fact store, narrated by a free-tier language model, and passed through a numeric guardrail that refuses any sentence carrying a number the fact store does not contain. Claims live in a version-controlled knowledge base with a status field (confirmed, retracted, refuted, pending, superseded), so a retraction is a state change, not a deletion.

Sources. knowledge/claims.json ident-001, ident-004, event-001, event-002, ident-008; results/CLAIMS_AUDIT.md; results/GTA_LINK_STAGE1.md; demo/showcase.html section c.

3. The corpus: repeat measurement across a manager change

Twelve Manchester United matches from the 2024-25 Premier League season form six home-and-away pairs against the same six opponents: Brighton, Crystal Palace, Liverpool, Southampton, Tottenham, and Fulham (single leg used for the demo). The pairing straddles the mid-season change of manager from Erik ten Hag to Ruben Amorim.

This design exists for one reason: measuring the same fixture twice is the only way to tell "Manchester United's style" apart from "what this specific opponent, on this specific day, made them do". A single match cannot answer the research question; a repeated pair can start to.

The demo built from this corpus is a pre-match opposition scouting pack for a fixture played twice. The Brighton pair is the primary demo, because Brighton's own manager stayed the same across both legs, making it a fair test of the design. The Southampton pair is the honest stress case: Southampton changed their own manager between legs, so the pack is run to the point where the evidence gate refuses, and what the system declines to say is itself the thesis being demonstrated.

Sources. docs/PROJECT_DEFINITION.md; results/MANUTD_IDENTITY_PROFILE.md.

4. Findings: what is stable, what is not

Some style dimensions repeat across the corpus. Others do not survive an opponent control. Both kinds of result are reported with equal weight.

4.1 Stable across the corpus

- **Possession control.** Repeat meetings correlate at $r = +0.83$. A pre-match forecast from one prior meeting reaches 7.92 percentage points mean absolute error, beating an uninformative baseline (10.63) and an Elo-based one (12.18).
- **The funnel shape (own third central, final third wide).** United play 46.9% of own-third ball time centrally and 23.9% of final-third ball time centrally, repeating in 11 of 12 matches (sign test $p = 0.0063$). It looks like a team trait, until Section 4.2.

4.2 Not stable, and the opponent control that killed the claim

The same funnel measurement, run on the opposition in the same twelve broadcasts, repeats in 12 of 12 matches ($p = 0.0005$) and finishes marginally *wider* than United do (22.6% central against United's 23.9%). It is Premier League geometry, not a Manchester United trait. The same control killed two further claims: a "rigid block" reading (compactness 6.17 +/- 0.39 m for United against 5.93 +/- 0.29 m for opponents, $F = 1.81$, $p = 0.34$) and a favoured-flank reading (5 of 12 matches lean left, $p = 0.7744$, no effect).

Defensive line height is also not stable. Leg-to-leg correlation $r = -0.40$: the home leg does not predict the away leg, and if anything predicts it backwards. Between-match standard deviation is 4.25 m on a mean of 28.3 m (coefficient of variation 15.0%).

Table 3. Stability summary

Dimension	Stable?	Evidence
Possession control	Yes	$r = +0.83$; forecast MAE 7.92pp beats baselines
Funnel shape (build central, finish wide)	No -- league trait	11/12 United, 12/12 opponents, opponents finish wider
Block compactness ("rigid" shape)	No -- league trait	United 6.17m vs opp 5.93m, $F=1.81$, $p=0.34$
Defensive line height	No	leg-to-leg $r = -0.40$, CV 15.0%
Favoured flank	No effect found	5/12 lean left, $p = 0.7744$

4.3 The manager change

Seven of nine players tracked in both eras stand higher up the pitch under Amorim (Garnacho +15.1 m being the largest mover). The team's own block-height metric does not cleanly separate the two managers (29.7 m under ten Hag to 26.9 m under Amorim, against a 4.25 m match-to-match standard deviation). The manager change is visible in the players, not measurably in the team shape, and the design confound must be stated: venue and manager flip together in all six pairs, so no number here separates "new manager" from "away instead of home".

Sample-size bound, stated as a limit. Team-versus-team rate comparisons (for example, "United press more than X") need 58-90 matches to separate two sides even with perfect tracking and zero measurement error, derived from published between-team and season variance. Twelve matches is five to eight times too few, before broadcast censoring is even considered. No improvement to the vision pipeline can rescue this; it is a sample-size bound.

Sources. results/MANUTD_IDENTITY_PROFILE.md sections 3-5; results/W1B_WINDOW_AND_SAMPLING.md; docs/PROJECT_DEFINITION.md.

5. The attribution research line

The question behind this line of work: can a passer, tackler or shooter be named, not just detected? The measured baseline before this research line: carrier attribution precision 0.846 (gate-hit) x 0.719 (team) x 0.609 (naming) = 0.350 (best variant 0.400).

5.1 Stage 1 -- tracklet repair (GTA-Link)

A training-free connector (Sun et al., ACCV 2024) merges fragmented tracklets by appearance embedding under spatio-temporal constraints. Measured on the SoccerNet-GSR public split:

Table 4. Stage 1 result

Metric	Before	After
GS-HOTA	22.85	24.18 (+1.33, 51/58 sequences helped)
Fragments per GT player	5.87	4.72

The connector helps tracking. It does not help naming. On 78 hand-judged carrier moments, naming went 0.609 to 0.750 -- one extra correct answer on a smaller denominator, not significant (Fisher $p = 0.35$). On a higher-power proxy (leave-one-group-out gallery top-1 over about 2,000 paired crops), the connector's default setting is flat ($p = 0.14$) and two stronger settings are significantly *worse* ($p = 0.0088$, $p = 0.0028$): at 70-80% merge precision, every wrongly merged crop becomes a permanent distractor. The negative Splitter half of GTA (reverses sign on the full run) was shipped off entirely.

5.2 Stage 2 -- the joint identity solver

A Lu-style constraint solve (identity constant within a tracklet, at most one player per frame, roster and substitution windows) was built with `scipy.optimize.milp`, no new dependency. Pre-declared gate, DEV/TEST split frozen in code before the single TEST run.

Gate 1, primary, pre-declared: FAIL. Solver versus appearance-only identity accuracy on TEST: +0.0045, Wilcoxon $p = 0.566$, 20 sequences helped, 18 hurt. The published jump this method is modelled on (50-55% to 85-89%) did not replicate here.

Measured cause: evidence starvation. Only 8.7% of tracklets carry any jersey read at all; an oracle restricted to that same evidence caps at 0.365. The published method that inspired this solve had dense external identity evidence from play-by-play text; jersey OCR alone does not supply that density. Constraints redistribute evidence between tracklets; they cannot create evidence that was never read.

The frontier finding. Per-row accuracy is blind to a precision/coverage trade: the baseline names 15.4% of rows at 88.2% precision, the frozen solver names 85.4% of rows at 26.4% precision, and per-row accuracy scores the two within 0.005 of each other. The solver traces the same precision/coverage curve the greedy baseline already sits on. It buys a dial, not a better frontier.

Gate 2 passes. GS-HOTA 23.37 to 24.19 on the same 38 held-out sequences, no regression. **The one clean win:** goalkeeper naming. Keeper slots filled go from 0 of 4 to 2 of 4 in all three labelled matches, correct starters every time; two centre-backs (de Ligt, van Dijk) enter the name set for the first time after zero mentions in 1,664 previously named tracks.

5.3 The evidence-density law

A simulator over FOOTPASS dense ground truth (97,397 action events, 54 matches), calibration- gated first: it reproduces Stage 2's own measured numbers (0.2604 predicted at 0.2576, appearance top-1 0.6366 predicted at 0.6413) with nothing fitted to those targets.

Table 5. The measured law: what attribution precision 0.85 at coverage ≥ 0.50 needs

Lever	Requirement	Multiple of today
OCR-like read density	$d = 0.347$ (from 0.087 today)	4.0x
Fragmentation reduction (alternative route)	~1 fragment / 30s on-screen	~7x less fragmentation
Read precision (weak lever)	-	1.3x

4x more reads and 7x less fragmentation are an exchange rate: either move closes the same gap.

Commentary as a naming channel: FAIL, retired by measurement. At its pre-declared gate (1.5 names/min, 0.60 precision, 4s lag IQR): coverage at the 0.85 precision floor is 0.000 across all 27 tested conditions. Measured cause: at a 4s lag, a commentary mention binds to the correct player 13.2% of the time, against a 12.5% chance rate (the ball moves about 24 m in 4 s). Even an oracle binder tops out at 0.295 coverage, below the bar. Stacked on real OCR it is net-harmful. The one-week build for this channel is cancelled for the price of a simulator run.

A hard ceiling, independent of any lever: 18.5% of labelled actions in FOOTPASS have an actor who is off-screen at the moment of the action. No pixel-only pipeline can exceed 0.81 coverage; the measured full-coverage arms of the simulator already sit at that ceiling.

Sources. results/GTA_LINK_STAGE1.md; results/IDENTITY_SOLVER_STAGE2.md; results/EVIDENCE_DENSITY_LAW.md; knowledge/claims.json gta-002, gta-004, ident-013 through ident-016.

6. Retractions and the correction trail as a feature

The knowledge base holds 102 logged claims: 72 confirmed, 12 retracted, 8 refuted by an external control, 8 pending, 2 superseded. A claim record carries its statement, status, method, sample size, caveats and evidence files. Retracting a claim changes its status; it does not delete the claim.

6.1 The calibration retraction

The B4 imputation model's 50%/90% predictive regions passed calibration 6 of 6 buckets on Metrica simulated censoring, and this was treated as a genuine contribution, printed on every frame of the tactical-clip demo. On real broadcast footage, measured against 4,900 liveness-guarded player re-appearances across four matches, the regions cover 21.2% where they promise 50% and 55.4% where they promise 90%. 14 of 14 testable buckets fail at both levels, with only 1.4-6.4 percentage points of spread between matches: not a one-match artefact.

The check that found this is itself a method contribution: when a tracked player walks back into frame after an occlusion, his re-appearance position is a measured answer to the prediction the model made while he was hidden. This is free ground truth on real footage, and it appears to be the first calibration check of an off-screen imputation model outside simulation in this literature; comparable models (DeepMind's Graph Imputer, a 2026 arXiv preprint, and our own B4) are all calibrated only in simulation.

What was ruled out as the sole cause: a measured observation-noise floor (0.541 m per axis) recovers the 50% level (55.0%) but leaves the 90% level 20 points short (70.4%); a displacement-matched Metrica subset still covers 52-59% and 92-95%, so selection bias does not explain the failure either.

6.2 Two other retractions on the same day

- **Face recognition as an identity source: closed by measurement.** Zero of 1,095 carrier moments across three matches produce a face reaching the roughly 50-pixel height recognition needs. Median face height is 9.9 px against a geometric ceiling of 16.8 px. This is an optics limit, not an engineering one.
- **Team mapping errors.** A standing screen (possession by kit cluster versus official majority, warning above 8 percentage points of disagreement) found two matches with flipped team labels. Four downstream claims -- the two mappings and two tactical reads built on them -- were marked retracted.

6.3 What the correction cost, and what it produced

The fix for the calibration retraction was not to soften the wording. The regions were measured against 4,900 real re-appearances, the failure was published, the claim was marked retracted in the knowledge base, and the tactical-clip renderer was changed so the measured numbers are burned into every frame beside the nominal ones. A self-check now asserts that no on-screen string can say "calibrated" without the word "NOT" attached. A separate rendering defect was found in the same pass: the clip renderer had only ever drawn dead track fragments as ghosts, never a genuinely occluded live player; with the fix, live-track occlusions drawn on the flagship passage went from 0% to 56%.

Sources. results/CALIBRATION_REPLICATION.md; results/FULL_MATCH_RECONSTRUCTION.md; knowledge/claims.json b4-014, b4-015; results/CROSS_MATCH_GALLERY.md; STATUS.md 2026-07-27 entry.

7. The SoccerNet FOOTPASS arena and the road to December

SoccerNet retired its Game State Reconstruction challenge in 2026 and replaced it with Player-Centric Ball Action Spotting: name the acting player by team and jersey number, scored Macro-F1, with an exact jersey match required. Its dataset, FOOTPASS, holds 54 matches, 81 hours of broadcast, and 102,992 validated (frame, team, jersey, action) tuples. Annotations are licensed CC BY-NC 4.0 on HuggingFace; video access requires a separate signed NDA.

This project has acquired the FOOTPASS annotations (97,397 visible events in train and validation, 54 of 54 games present) under the narrow exception recorded in project rules: an NDA personally signed by the project owner. Video access is a separate NDA decision, not yet made, and is required before FOOTPASS can grade this pipeline end to end from raw pixels.

Two facts from a survey of the current SoccerNet leaderboard sharpen the plan. First, across the whole GSR 2025 leaderboard, tracklet-association scores (60-85) far exceed identity scores (14-51); the leaderboard winner states its own failures were "mostly jersey number recognition". This matches this project's own measured result: the connector repairs tracking, not naming. Second, no winning team's code is public across either GSR year, so the 4th-place recipe (a stack close to ours, plus a stronger detector, pose-cropped jersey OCR, and a geometric goalkeeper heuristic) is the most useful reference point available.

The re-ranked build order, in the order the evidence-density law justifies: (1) OCR densification toward the measured 4x target -- persisting per-crop reads, pose or torso crops, a small vision-language-model zoom trial, and solving over longer windows; (2) fragmentation reduction toward the measured ~1-per-30s target, since a perfect within-chunk association already reaches 0.505 coverage on today's read volume; (3) one end-to-end from-pixels run on the FOOTPASS validation games once (1) and (2) have moved. Commentary is retired to report colour only, never attribution evidence.

Sources. docs/ATTRIBUTION_RESEARCH_PLAN.md post-survey revision; STATUS.md 2026-07-28 entries; docs/SOCCERNET_SURVEY_RAW.md.